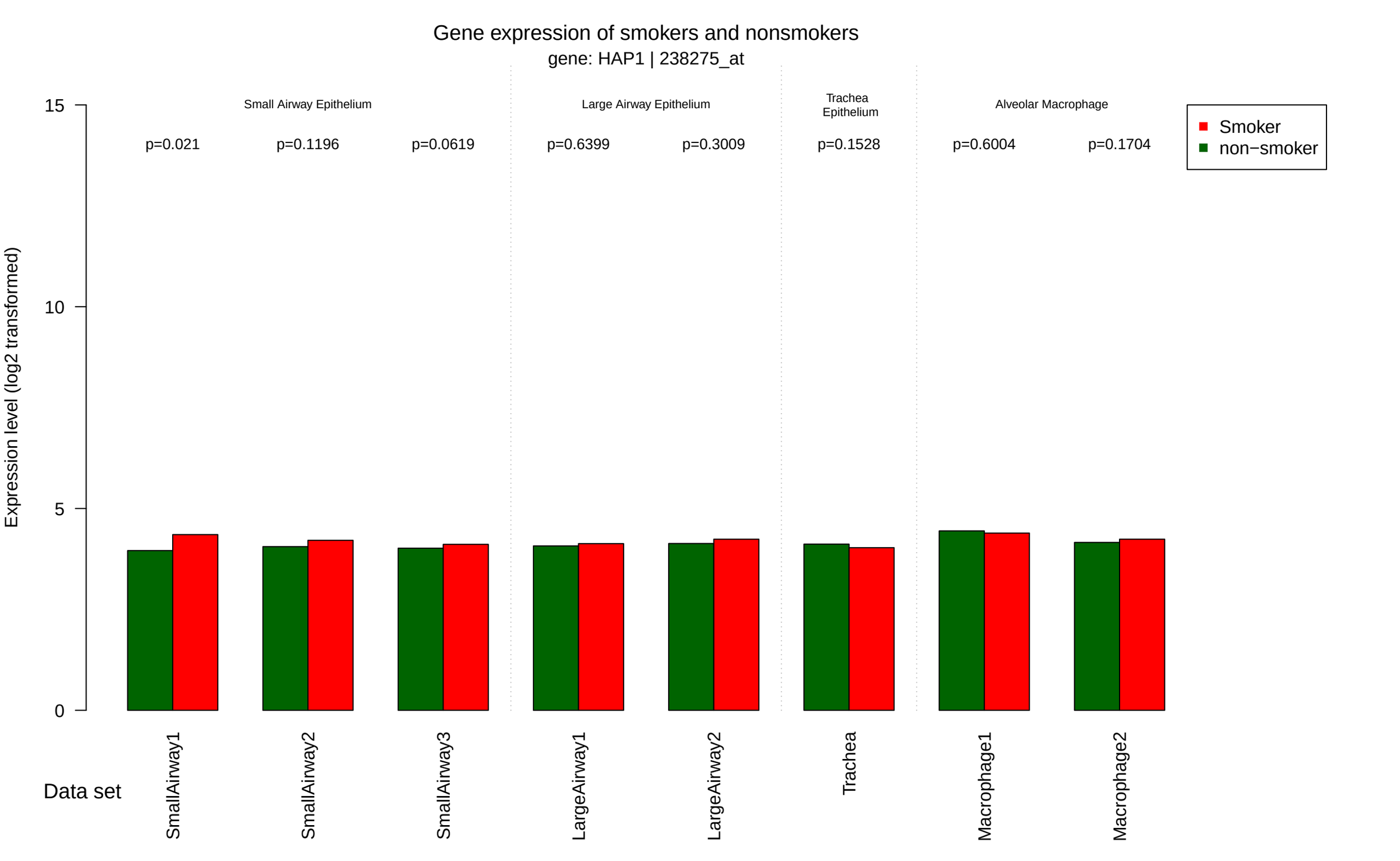
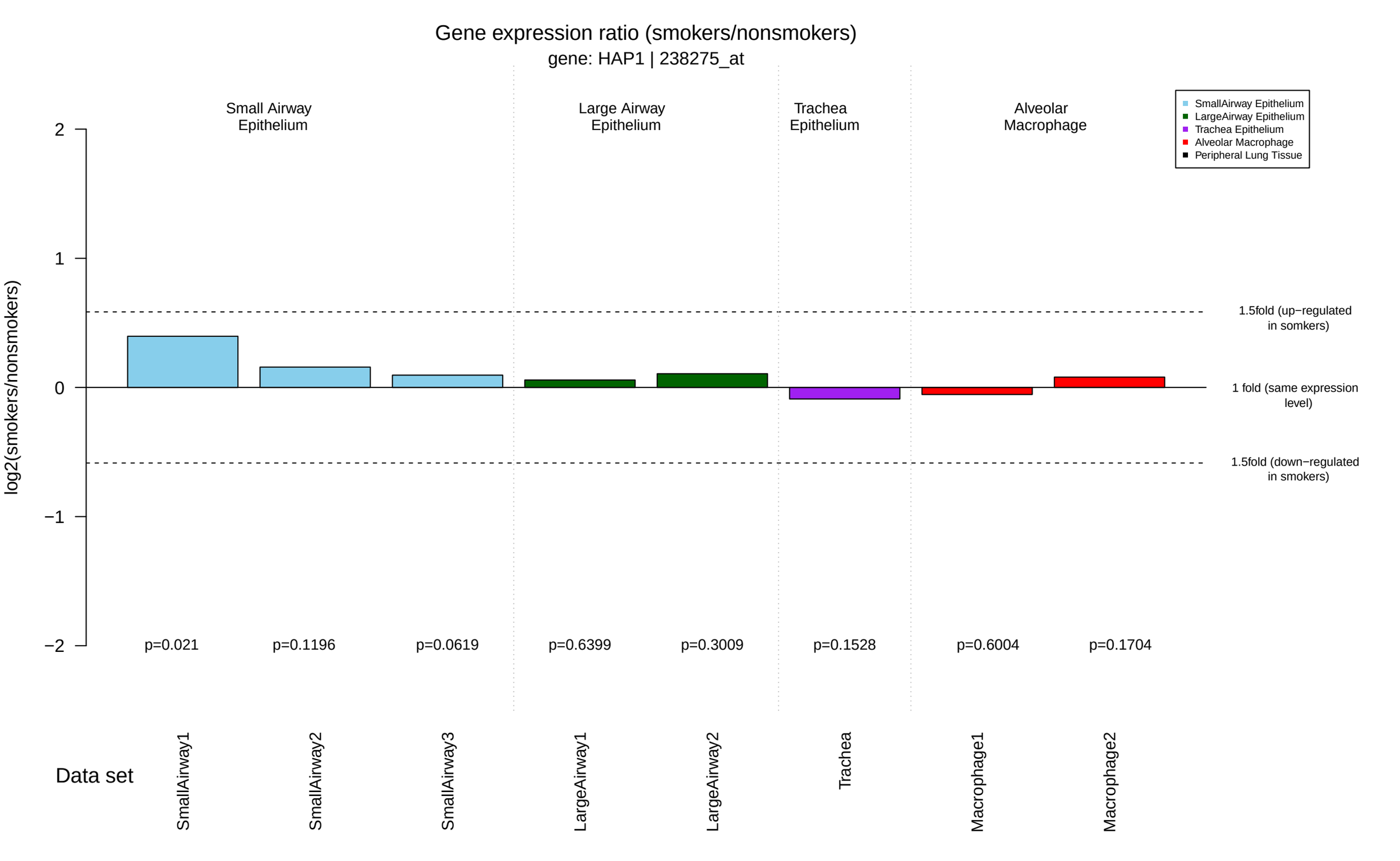
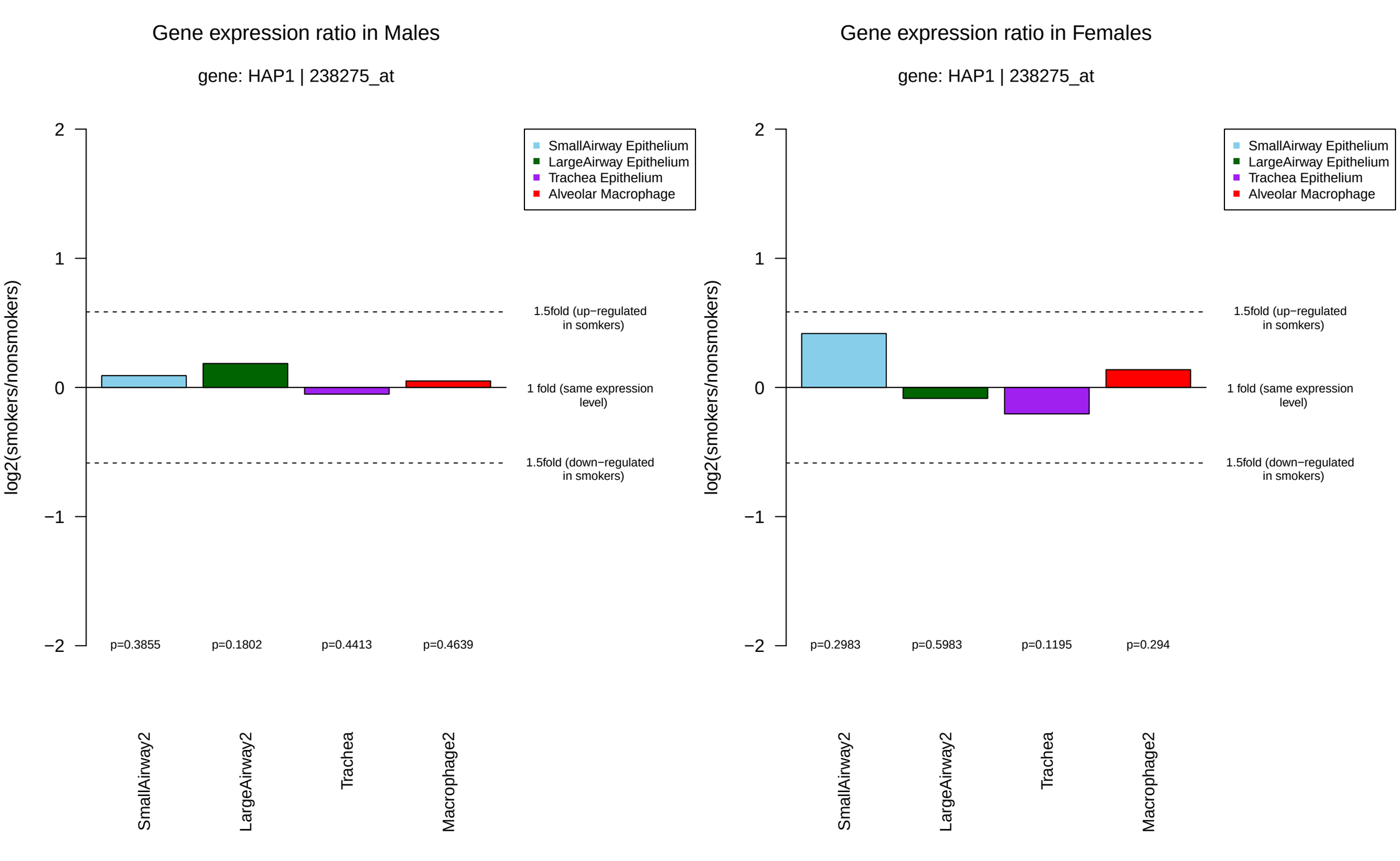
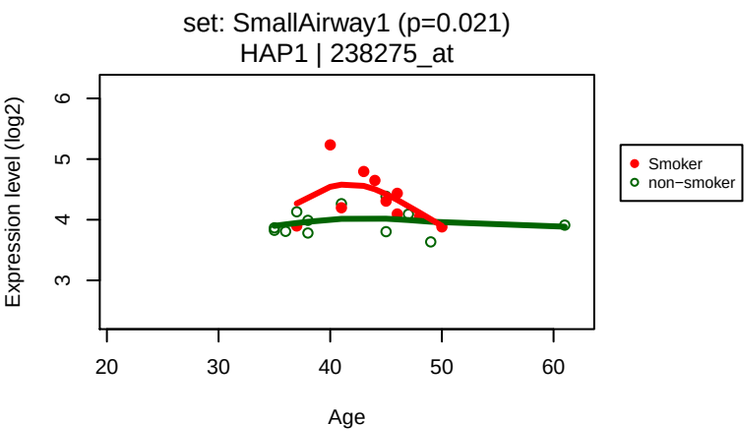
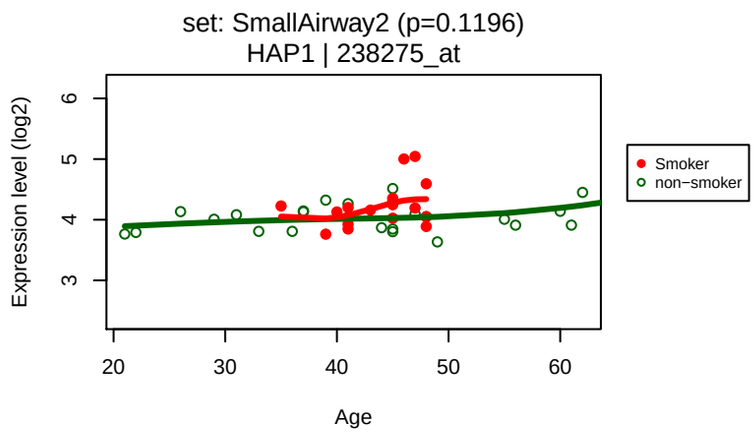
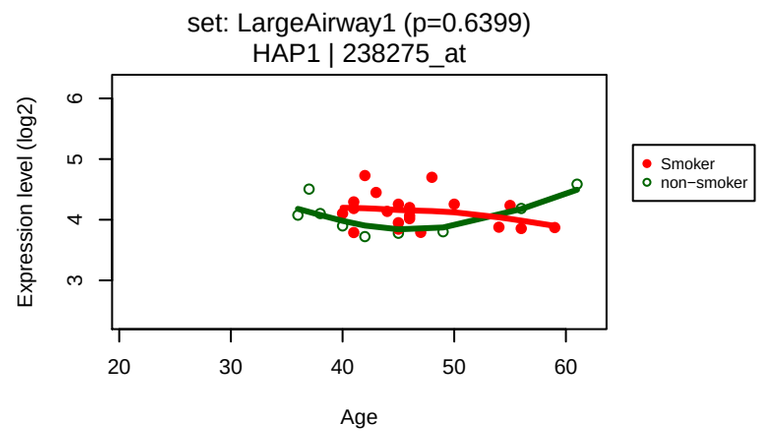

Gene expression of smokers and nonsmokers
gene: HAP1 | 238275_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.021
p=0.1196
p=0.0619
p=0.6399
p=0.3009
p=0.1528
p=0.6004
p=0.1704
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

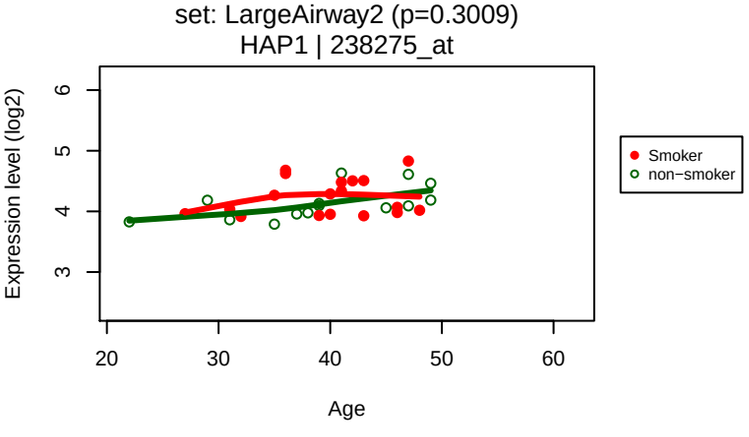

Gene expression ratio (smokers/nonsmokers)
gene: HAP1 | 238275_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.021
p=0.1196
p=0.0619
p=0.6399
p=0.3009
p=0.1528
p=0.6004
p=0.1704
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

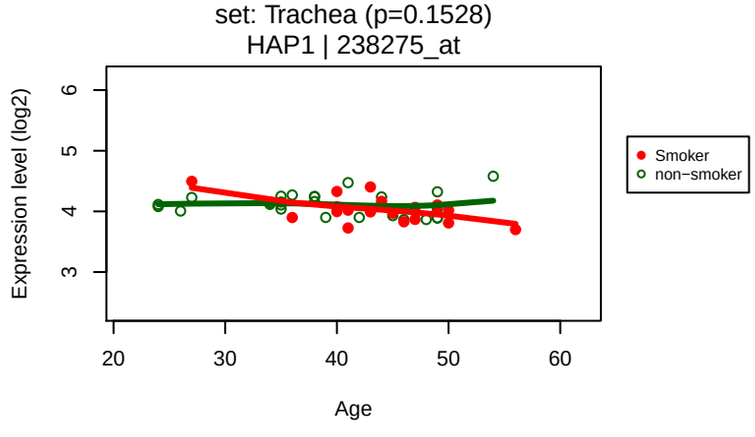

Gene expression ratio in Males
gene: HAP1 | 238275_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3855
p=0.1802
p=0.4413
p=0.4639
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HAP1 | 238275_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2983
p=0.5983
p=0.1195
p=0.294
SmallAirway2
LargeAirway2
Trachea
Macrophage2

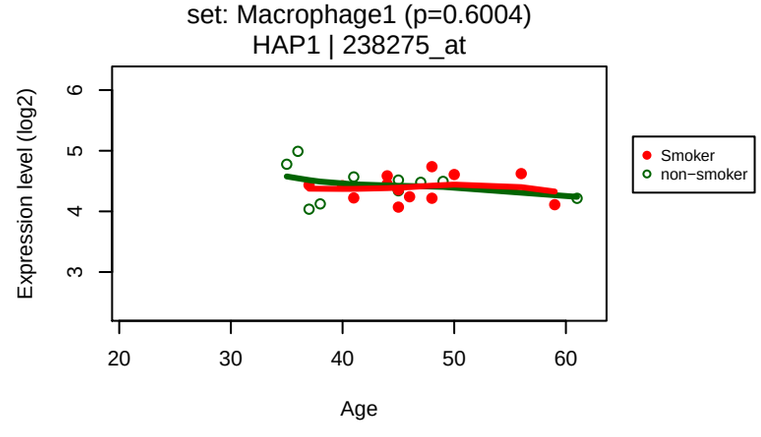

set: SmallAirway1 (p=0.021)
HAP1 | 238275_at
Expression level (log2)
Age
Smoker
non−smoker

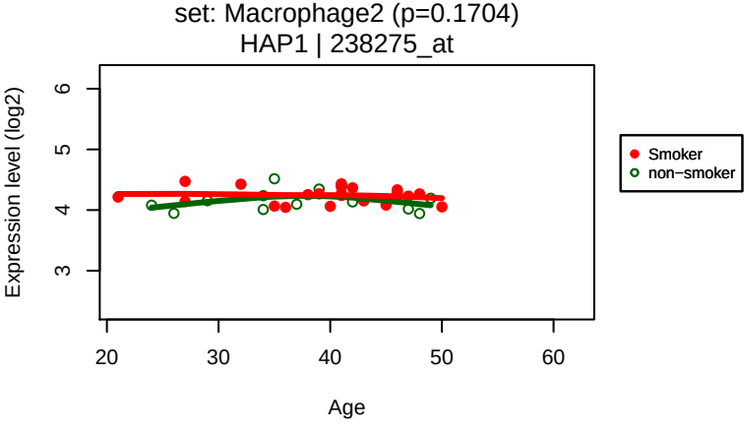

set: SmallAirway2 (p=0.1196)
HAP1 | 238275_at
Expression level (log2)
Age
Smoker
non−smoker

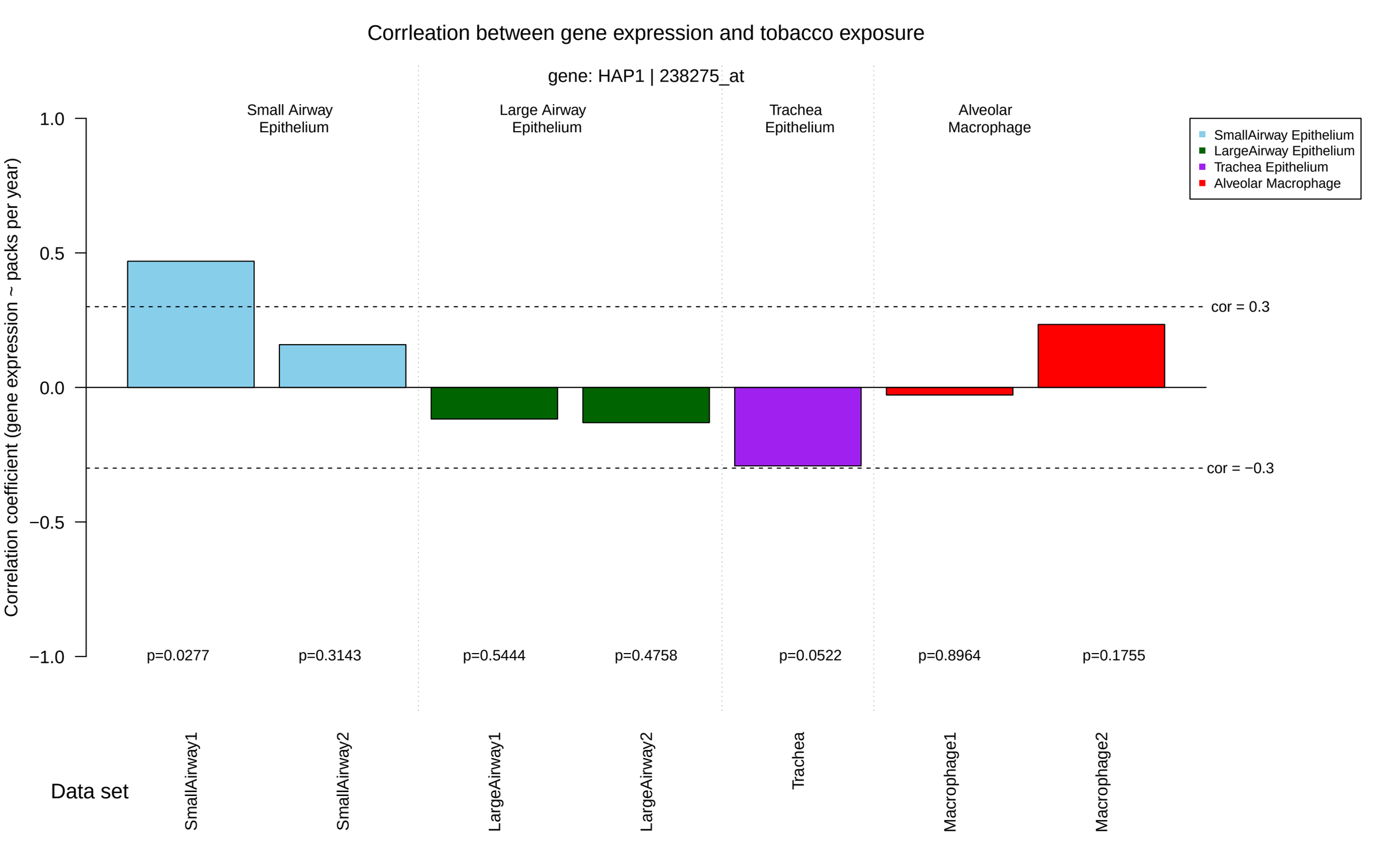

set: LargeAirway1 (p=0.6399)
HAP1 | 238275_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3009)
HAP1 | 238275_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1528)
HAP1 | 238275_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6004)
HAP1 | 238275_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1704)
HAP1 | 238275_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HAP1 | 238275_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0277
p=0.3143
p=0.5444
p=0.4758
p=0.0522
p=0.8964
p=0.1755
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2